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Inventor(s)	Glennon; Kevin James et al.

Methods and apparatus for automating shutdown processes of gas grills

Abstract

Example methods and apparatus for automating shutdown processes of gas grills are disclosed. An example grill includes a first burner, a second burner, a first burner valve operatively positioned between the first burner and a manifold of the grill, and a second burner valve operatively positioned between the second burner and the manifold. The example grill further includes a controller operatively coupled to the first burner valve and the second burner valve. In response to determining that a shutdown request has been received at the grill, the controller is to instruct the first burner valve and the second burner valve to close.

Inventors: Glennon; Kevin James (Hoffman Estates, IL), Mecker; William Alexander (Kildeer, IL)

Applicant: Weber-Stephen Products LLC (Palatine, IL)

Family ID: 85480034

Assignee: Weber-Stephen Products LLC (Palatine, IL)

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Primary Examiner: Savani; Avinash A

Attorney, Agent or Firm: Avantech Law

Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/242,901, filed Sep. 10, 2021. The entirety of U.S. Provisional Patent Application No. 63/242,901 is hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to gas grills and, more specifically, to methods and apparatus for automating shutdown processes of gas grills.

BACKGROUND

(2) The process of shutting down a conventional gas grill includes a high degree of user (e.g., human) interaction. This is particularly true with regard to multi-burner gas grills. For example, in a conventional gas grill having two burners (e.g., a left burner tube and a right burner tube), the shutdown process typically requires a user to (1) manually push in and rotate a first control knob operatively coupled to a first valve associated with a first one of the two burners, thereby closing off a flow of gas from a manifold of the grill into the first one of the two burners, (2) manually push in and rotate a second control knob operatively coupled to a second valve associated with a second one of the two burners, thereby closing off a flow of gas from the manifold into the second one of the two burners, and (3) manually rotate a knob or lever operatively coupled to a valve of a fuel source (e.g., a valve of a fuel tank or a valve of a natural gas line), thereby closing off a flow of gas from the fuel source into the manifold.

(3) Although relatively simple to perform, the shutdown process for the above-described two-burner conventional gas grill nonetheless involves three separate steps that must be manually performed by a user of the grill according to the specified sequence. In a gas grill having more than two burners (e.g., three burners, four burners, five burners, etc.), the shutdown process becomes even more involved in that the user is required to repeat the above-described second step of the shutdown process for each additional burner of the grill. The shutdown process can accordingly become cumbersome, providing for a user experience that is less than ideal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an example grill constructed in accordance with the teachings of this disclosure.

(2) FIG. 2 is a perspective view of an example cookbox constructed in accordance with the teachings of this disclosure.

(3) FIG. 3 is a front view of an example user interface constructed in accordance with the teachings of this disclosure.

(4) FIG. 4 is a flowchart representative of example machine-readable instructions and/or example operations that may be executed by processor circuitry to implement an automated shutdown

process via the grill of FIG. 1.

(5) FIG. 5 is a flowchart representative of example machine-readable instructions and/or example operations that may be executed by processor circuitry to implement another automated shutdown process via the grill of FIG. 1.

(6) FIG. 6 is a flowchart representative of example machine-readable instructions and/or example operations that may be executed by processor circuitry to implement another automated shutdown process via the grill of FIG. 1.

(7) FIG. 7 is a flowchart representative of example machine-readable instructions and/or example operations that may be executed by processor circuitry to implement another automated shutdown process via the grill of FIG. 1.

(8) FIG. 8 is a block diagram of an example processor platform including processor circuitry structured to execute and/or instantiate the machine-readable instructions and/or operations of FIGS. 4-7 to implement the grill of FIG. 1.

(9) FIG. 9 is a block diagram of an example implementation of the processor circuitry of FIG. 8.

(10) FIG. 10 is a block diagram of another example implementation of the processor circuitry of FIG. 8.

(11) Certain examples are shown in the above-identified figures and described in detail below. In describing these examples, like or identical reference numbers are used to identify the same or similar elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale or in schematic for clarity and/or conciseness.

(12) Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.

DETAILED DESCRIPTION

(13) The process of shutting down a conventional gas grill includes a high degree of user (e.g., human) interaction. This is particularly true with regard to multi-burner gas grills. For example, in a conventional gas grill having two burners (e.g., a left burner tube and a right burner tube), the shutdown process typically begins with a user manually pushing in and then manually rotating a first control knob that is located along a control panel of the grill and operatively coupled to a first valve associated with a first one of the two burners of the grill (e.g., a control knob for the left burner tube). The acts of pushing in and subsequently rotating the first control knob cause the first valve to close, thereby closing off a flow of gas from a manifold of the grill into the first one of the two burners. A second step in the shutdown process typically requires the user to manually push in and then manually rotate a second control knob that is located along the control panel of the grill and operatively coupled to a second valve associated with a second one of the two burners of the grill (e.g., a control knob for the right burner tube). The acts of pushing in and subsequently rotating the second control knob cause the second valve to close, thereby closing off a flow of gas from the manifold into the second one of the two burners.

(14) A third step in the shutdown process typically requires the user to manually rotate a knob of a fuel tank that is operatively coupled to a valve of the fuel tank, with the fuel tank being connected to the manifold of the grill via a conduit. The act of rotating the knob of the fuel tank causes the valve of the fuel tank to close, thereby closing off a flow of gas from the fuel tank, through the conduit, and into the manifold. In some instances, this third step of the shutdown process may additionally require the user to open a door of a cabinet of the grill in order to access the knob of the fuel tank, and to thereafter close the door of the cabinet once the valve of the fuel tank has been

closed. In some implementations, the grill may instead be connected to a natural gas line rather than a fuel tank. In such examples, the above-described third step of the shutdown process may instead require the user to manually rotate a knob or a lever that is operatively coupled to a safety valve located in a conduit that connects the natural gas line to the manifold of the grill.

(15) Although relatively simple to perform, the shutdown process for the above-described two-burner conventional gas grill nonetheless involves at least three separate steps that must be manually performed by a user of the grill according to the specified sequence. In a gas grill having more than two burners (e.g., three burners, four burners, five burners, etc.), the shutdown process becomes even more involved in that the user is required to repeat the above-described second step of the shutdown process for each additional burner of the grill. The shutdown process can accordingly become cumbersome, providing for a user experience that is less than ideal.

(16) Relative to the known shutdown processes described above that require a high degree (e.g., a high number) of user (e.g., human) interaction(s) to shutdown multiple (e.g., all) burners of a gas grill, the automated shutdown processes disclosed herein advantageously require a reduced degree (e.g., a reduced number) of user interaction(s) to shutdown multiple (e.g., all) burners of a gas grill. In some examples, automated shutdown processes disclosed herein require as few as two user interactions to shutdown multiple (e.g., all) burners of a gas grill. In other examples, automated shutdown processes disclosed herein require as few as one user interaction to shutdown multiple (e.g., all) burners of a gas grill. Implementing the automated shutdown processes disclosed herein to reduce the degree (e.g., the number) of user interaction(s) required to shutdown multiple (e.g., all) burners of a gas grill advantageously makes the shutdown process less cumbersome in terms of the extent of user involvement that is required in connection with the shutdown process. The automated shutdown processes disclosed herein thereby provide a user experience that is improved relative to that provided by known shutdown processes for gas grills.

(17) The above-identified features as well as other advantageous features of example methods and apparatus for automating shutdown processes of gas grills as disclosed herein are further described below in connection with the figures of the application. As used herein in a mechanical context, the term “configured” means sized, shaped, arranged, structured, oriented, positioned, and/or located. For example, in the context of a first object configured to fit within a second object, the first object is sized, shaped, arranged, structured, oriented, positioned, and/or located to fit within the second object. As used herein in an electrical and/or computing context, the term “configured” means arranged, structured, and/or programmed. For example, in the context of a controller configured to perform a specified operation, the controller is arranged, structured, and/or programmed (e.g., based on machine-readable instructions) to perform the specified operation. As used herein, the phrase “in electrical communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events. As used herein, the term “processor circuitry” is defined to include (i) one or more special purpose electrical circuits structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmed with instructions to perform specific operations and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of processor circuitry include programmed microprocessors, Field Programmable Gate Arrays (FPGAs) that may instantiate instructions, Central Processor Units (CPUs), Graphics Processor Units (GPUs), Digital Signal Processors (DSPs), XPU, or microcontrollers and integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of processor circuitry (e.g., one or more FPGAs, one or more

CPUs, one or more GPUs, one or more DSPs, etc. and/or a combination thereof) and application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of the processing circuitry is/are best suited to execute the computing task(s).

(18) FIG. 1 is a block diagram of an example grill **100** constructed in accordance with the teachings of this disclosure. The grill **100** of FIG. 1 is a gas grill including a plurality of burners. In the illustrated example of FIG. 1, the grill **100** includes an example first burner **102** and an example second burner **104**. In other examples, the grill **100** can include one or more other burner(s) (e.g., a third burner, a fourth burner, a fifth burner, etc.) in addition to the first burner **102** and the second burner **104** shown and described in connection with FIG. 1. The first burner **102** and the second burner **104** of FIG. 1 are each constructed as a burner tube (e.g., a linear burner tube) including a gas inlet for receiving a flow of combustible gas, and further including a plurality of apertures configured to emit flames generated in response to ignition of the gas flowing into and/or through the burner tube.

(19) The grill **100** further includes a cookbox (not shown in FIG. 1) that supports, carries, and/or houses the burners (e.g., the first burner **102** and the second burner **104**) of the grill **100**, with respective ones of the burners being spaced apart from one another within the cookbox. FIG. 2 is a perspective view of an example cookbox **200** constructed in accordance with the teachings of this disclosure. As shown in FIG. 2, the cookbox **200** supports, carries, and/or houses a total of five example burners **202** (e.g., including the first burner **102** and the second burner **104** of FIG. 1), with each of the five burners **202** being spaced apart from one another within the cookbox **200**. In other examples, the cookbox **200** can support, carry, and/or house a different number (e.g., two, three, four, six, etc.) of burners **202**. In the illustrated example of FIG. 2, each of the burners **202** is constructed as a linear burner tube positioned in a front-to-rear orientation within the cookbox **200** (e.g., extending from a front wall **204** of the cookbox **200** to a rear wall **206** of the cookbox **200**). In other examples, one or more of the burner(s) **202** can have a different shape (e.g., a non-linear shape such as a P-tube), and/or can have a different orientation (e.g., a left-to-right orientation) within the cookbox **200**. It should accordingly be understood that the cookbox configuration shown in FIG. 2 is but one example of a cookbox **200** that can be implemented as part of the grill **100** of FIG. 1. The grill **100** of FIG. 1 further includes a frame and/or a cabinet for supporting the cookbox of the grill **100**, and a lid for covering and/or enclosing the cookbox of the grill **100**. The grill **100** of FIG. 1 further includes one or more side table(s) coupled to the frame, to the cabinet, and/or to the cookbox of the grill **100**. Various components of the grill **100** of FIG. 1 described herein can be supported by, carried by, housed by, mounted to, and/or otherwise coupled to at least one of the cookbox, the lid, the frame, the cabinet, or the side table(s) of the grill **100**.

(20) In addition to the first burner **102** and the second burner **104** described above, the grill **100** of FIG. 1 further includes an example fuel source **106**, an example fuel source valve **108**, an example manifold **110**, an example first burner valve **112**, an example second burner valve **114**, an example first ignitor **116**, an example second ignitor **118**, an example first encoder **120**, an example first control knob **122**, an example second encoder **124**, an example second control knob **126**, an example user interface **128** (e.g., including one or more example input device(s) **130** and one or more example output device(s) **132**), an example network interface **134** (e.g., including one or more example communication device(s) **136**), an example controller **138** (e.g., including example valve control circuitry **140** and example flame detection circuitry **142**), and an example memory **144**. The grill **100** of FIG. 1 is configured to communicate (e.g., wirelessly communicate) with one or more example remote device(s) **146**, as further described below.

(21) The grill **100** of FIG. 1 includes a control system for implementing one or more automated shutdown process(es) that rely on a minimal number of user (e.g., human) interactions with the grill **100** in connection with extinguishing multiple ones (e.g., all) of the burners of grill **100**. In the illustrated example of FIG. 1, the control system of the grill **100** includes the fuel source valve **108**, the first burner valve **112**, the second burner valve **114**, the first ignitor **116**, the second ignitor **118**,

the first encoder **120**, the second encoder **124**, the user interface **128** (e.g., including the input device(s) **130** and the output device(s) **132**), the network interface **134** (e.g., including the communication device(s) **136**), the controller **138** (e.g., including the valve control circuitry **140** and the flame detection circuitry **142**), and the memory **144**. In other examples, one or more of the aforementioned components of the grill **100** can be omitted from the control system of the grill **100**. For example, the fuel source valve **108** can be omitted from the control system of the grill **100** in instances where the fuel source valve **108** is not configured to be electrically controlled and/or electrically actuated by the controller **138**, with the fuel source valve **108** instead being configured only for manual control and/or manual actuation. As another example, the first encoder **120** and the second encoder **124** can be omitted from the control system of the grill **100**, with certain functional and/or operational aspects of the first encoder **120** and the second encoder **124** in some instances being alternatively performed by one or more button(s), dial(s), knob(s), switch(es), touchscreen(s), and/or other input device(s) **130** of the user interface **128** of the grill **100**. In still other examples, the control system of the grill **100** can further include the remote device(s) **146** that are configured to communicate (e.g., wirelessly communicate) with the grill **100**.

(22) The control system of the grill **100** of FIG. **1** is powered and/or operated by a power source. For example, the electrical components that form the control system of the grill **100** can be powered and/or operated by DC power supplied via one or more on-board or connected batteries of the grill **100**. As another example, the electrical components that form the control system of the grill **100** can alternatively be powered and/or operated by AC power supplied via household electricity or wall power to which the grill **100** is connected. The grill **100** includes a power button (e.g., a power switch) that is configured to enable (e.g., power on) or disable (e.g., power off) the control system of the grill **100** in response to the power button being manually actuated by a user of the grill **100**.

(23) In some examples, one or more component(s) (e.g., the user interface **128**, the controller **138**, etc.) of the control system of the grill **100** is/are configured to transition from an active (e.g., awake) state to an inactive (e.g., sleep) state while the component(s) and/or, more generally, the control system is/are powered on. In this regard, the inactive (e.g., sleep) state is a low power state in which the control system consumes less power relative to the amount of power consumed by the control system when operating in the active (e.g., awake) state. In some examples, the control system of the grill **100** is configured to automatically transition from the active state to the inactive state in response to the passage of a threshold period of time in which one or more specified component(s) of the control system does/do not receive an input and/or an instruction that is indicative of a user request and/or command. In some examples, the control system of the grill **100** is configured to automatically transition from the inactive state to the active state in response to one or more specified component(s) of the control system receiving an input and/or an instruction that is indicative of a user request and/or command (e.g., a wake request), as further described below.

(24) The grill **100** of FIG. **1** further includes an example gas train **148** that extends from the fuel source **106** to the manifold **110** of the grill **100**, and from the manifold **110** to respective ones of the first burner **102** and the second burner **104** of the grill **100**. The gas train **148** can be implemented via one or more conduit(s) (e.g., one or more rigid or flexible pipe(s), tube(s), etc.) that are configured to carry combustible gas from the fuel source **106** to the first burner **102** and/or the second burner **104** of the grill **100**. In some examples, the fuel source **106** is implemented as a fuel tank (e.g., a propane tank) containing combustible gas. In such examples, the fuel source **106** will typically be located partially or fully within the cabinet of the grill, partially or fully within a spatial footprint formed by the frame of the grill, below the cookbox of the grill and partially or fully within a spatial footprint formed by the cookbox of the grill, or below the cookbox of the grill and partially or fully within a spatial footprint formed by one of the side tables of the grill. In other examples, the fuel source **106** can instead be implemented as a piped (e.g., household) natural gas line that provides an accessible flow of combustible gas.

(25) The fuel source valve **108** of FIG. **1** is coupled to and operatively positioned within the gas train **148** between the fuel source **106** and the manifold **110** of the grill **100**. The fuel source valve **108** is configured to be movable between an open position that enables gas contained within the fuel source **106** to flow from the fuel source **106** into the manifold **110**, and a closed position that prevents gas contained within the fuel source **106** from flowing into the manifold **110**. In the illustrated example of FIG. **1**, the fuel source valve **108** is operatively coupled to (e.g., in electrical communication with) the controller **138** of the grill **100**, with the fuel source valve **108** being implemented as a controllable electric valve (e.g., a solenoid valve) that is configured to transition from the open position to the closed position, and vice-versa, in response to instructions, commands, and/or signals (e.g., a supply of current) generated by the controller **138**. In other examples, the fuel source valve **108** can instead be implemented as a valve having a knob or a lever operatively coupled (e.g., mechanically coupled) thereto, with the knob or the lever being configured to be electrically actuated (e.g., via a motor) in response to instructions, commands, and/or signals generated by the controller **138** of the grill **100**. In still other examples, the fuel source valve **108** may have no electrically-controllable components, in which case actuation of the fuel source valve **108** from the open position to the closed position, and vice-versa, occurs in response to a user of the grill **100** manually actuating a knob or a lever that is operatively coupled (e.g., mechanically coupled) to the fuel source valve **108**.

(26) The first burner valve **112** of FIG. **1** is coupled to and operatively positioned within the gas train **148** between the manifold **110** and the first burner **102** of the grill **100**. In some examples, a gas inlet of the first burner valve **112** is located within the manifold **110**, and a gas outlet of the first burner valve **112** is located within the first burner **102**. The first burner valve **112** is configured to be movable between an open position that enables gas contained within the manifold **110** to flow from the manifold **110** into the first burner **102**, and a closed position that prevents gas contained within the manifold **110** from flowing into the first burner **102**. In the illustrated example of FIG. **1**, the first burner valve **112** is operatively coupled to (e.g., in electrical communication with) the controller **138** of the grill **100**, with the first burner valve **112** being implemented as a controllable electric valve (e.g., a solenoid valve) that is configured to transition from the open position to the closed position, and vice-versa, in response to instructions, commands, and/or signals (e.g., a supply of current) generated by the controller **138**. In some examples, the first burner valve **112** is controllable to any position (e.g., infinite position control) between the above-described open position (e.g., fully open) and the above-described closed position (e.g., fully closed). Unlike known burner valves of conventional gas grills, the first burner valve **112** does not include a stem that is mechanically coupled to a user-accessible control knob of the grill, whereby the control knob traditionally facilitates manual control and/or manual actuation of the operable position of the burner valve. The first burner valve **112** of FIG. **1** is accordingly only controllable and/or actuatable via the “control-by-wire” functionality referenced above and further described herein.

(27) The second burner valve **114** of FIG. **1** is coupled to and operatively positioned within the gas train **148** between the manifold **110** and the second burner **104** of the grill **100**. In some examples, a gas inlet of the second burner valve **114** is located within the manifold **110**, and a gas outlet of the second burner valve **114** is located within the second burner **104**. The second burner valve **114** is configured to be movable between an open position that enables gas contained within the manifold **110** to flow from the manifold **110** into the second burner **104**, and a closed position that prevents gas contained within the manifold **110** from flowing into the second burner **104**. In the illustrated example of FIG. **1**, the second burner valve **114** is operatively coupled to (e.g., in electrical communication with) the controller **138** of the grill **100**, with the second burner valve **114** being implemented as a controllable electric valve (e.g., a solenoid valve) that is configured to transition from the open position to the closed position, and vice-versa, in response to instructions, commands, and/or signals (e.g., a supply of current) generated by the controller **138**. In some

examples, the second burner valve **114** is controllable to any position (e.g., infinite position control) between the above-described open position (e.g., fully open) and the above-described closed position (e.g., fully closed). Unlike known burner valves of conventional gas grills, the second burner valve **114** does not include a stem that is mechanically coupled to a user-accessible control knob of the grill, whereby the control knob traditionally facilitates manual control and/or manual actuation of the operable position of the burner valve. The second burner valve **114** of FIG. **1** is accordingly only controllable and/or actuatable via the “control-by-wire” functionality referenced above and further described herein.

(28) The first ignitor **116** of FIG. **1** is mechanically coupled and/or operatively positioned relative to the first burner **102** of the grill **100**. More specifically, the first ignitor **116** is located adjacent the first burner **102** at a position that enables the first ignitor **116** to ignite combustible gas as the gas emanates from within the first burner **102** via apertures formed in the first burner **102**. The first ignitor **116** of FIG. **1** is operatively coupled to (e.g., in electrical communication with) the controller **138** of the grill **100**, with the first ignitor **116** being configured to generate sparks (e.g., via a spark electrode of the first ignitor **116**) and/or otherwise induces ignition of the combustible gas in response to an instruction, a command, and/or a signal generated by the controller **138**.

(29) The second ignitor **118** of FIG. **1** is mechanically coupled and/or operatively positioned relative to the second burner **104** of the grill **100**. More specifically, the second ignitor **118** is located adjacent the second burner **104** at a position that enables the second ignitor **118** to ignite combustible gas as the gas emanates from within the second burner **104** via apertures formed in the second burner **104**. The second ignitor **118** of FIG. **1** is operatively coupled to (e.g., in electrical communication with) the controller **138** of the grill **100**, with the second ignitor **118** being configured to generate sparks (e.g., via a spark electrode of the second ignitor **118**) and/or otherwise induces ignition of the combustible gas in response to an instruction, a command, and/or a signal generated by the controller **138**.

(30) In some examples, the first ignitor **116** and/or the second ignitor **118** of FIG. **1** can respectively be structured, configured, and/or implemented as one of the various ignitors described in U.S. patent application Ser. No. 17/144,038, filed on Jan. 7, 2021. In such examples, the first ignitor **116** and/or the second ignitor **118** of FIG. **1** can respectively be mechanically coupled to a corresponding one of the first burner **102** and/or the second burner **104** of the grill **100** via a ceramic harness as described in U.S. patent application Ser. No. 17/144,038. The entirety of U.S. patent application Ser. No. 17/144,038 is hereby incorporated by reference herein.

(31) The first encoder **120** of FIG. **1** is mechanically coupled to the first control knob **122** of FIG. **1** and operatively coupled to (e.g., in electrical communication with) the controller **138** of FIG. **1**. In this regard, the first encoder **120** of FIG. **1** is implemented as a rotary encoder having a rotatable shaft to which the first control knob **122** is mechanically coupled. The rotatable shaft of the first encoder **120** can be rotated relative to a fixed portion of the first encoder **120** via user interaction with the first control knob **122** (e.g., manual rotation of the first control knob **122**). The fixed portion of the first encoder **120** includes one or more sensor(s) that is/are configured to sense, measure, and/or detect the relative angular position of the rotatable shaft and/or the relative angular position of the first control knob **122**. Data, information, and/or signals that is/are sensed, measured, and/or detected by the sensor(s) of the first encoder **120** can be transmitted directly to the controller **138** of FIG. **1**, and/or can be transmitted to and stored in the memory **144** of FIG. **1**. In some examples, the sensor(s) of the first encoder **120** is/are further configured to sense, measure, and/or detect a translational movement of the rotatable shaft relative to the fixed portion of the first encoder **120**, as may occur in response to a user of the grill **100** pushing or pressing on the first control knob **122** in a direction that is generally perpendicular to the direction(s) in which the first control knob **122** is configured to be rotated by the user.

(32) In some examples, the first encoder **120** of FIG. **1** is mounted to a control panel of the grill **100** (e.g., located along the front wall **204** of the cookbox **200** of the grill **100**) such that the first

encoder **120** is located at a position on the control panel that would conventionally be occupied by a stem of a burner valve that corresponds to the first burner valve **112** of FIG. **1**. Such an example further facilitates locating the first control knob **122** of FIG. **1** at a position on or along the control panel that would conventionally be occupied by a control knob that is mechanically coupled to the stem of the burner valve that corresponds to the first burner valve **112** of FIG. **1**. While the first control knob **122** of FIG. **1** may accordingly be located at a position on or along the control panel of the grill **100** that mimics the position at which a traditional control knob is located, user actuation (e.g., manual rotation) of the first control knob **122** of FIG. **1** provides a response that differs greatly from that provided by user actuation (e.g., manual rotation) of a traditional control knob.

(33) For example, conventional multi-burner gas grills typically include a plurality of control knobs (e.g., located on or along a control panel of the grill), with each control knob being physically associated with a corresponding one of the burners of the gas grill by virtue of (1) a first mechanical connection existing between the control knob and a stem of a corresponding burner valve (e.g., such that rotation of the control knob by a user of the grill opens, closes, or otherwise adjusts the position of the burner valve), and (2) a second mechanical connection existing between the burner valve and the corresponding burner. By contrast, the grill **100** of FIG. **1** implements a “control-by-wire” architecture that eliminates the first of the aforementioned mechanical connections in favor of (1) a mechanical connection existing between the first control knob **122** of FIG. **1** and the first encoder **120** of FIG. **1**, (2) a first electrical connection existing between the first encoder **120** of FIG. **1** and the controller **138** and/or the memory **144** of FIG. **1**, and (3) a second electrical connection existing between the controller **138** of FIG. **1** and the first burner valve **112** of FIG. **1**.

(34) Although the first control knob **122** of FIG. **1** is not mechanically coupled to the first burner valve **112** of FIG. **1**, rotation of the first control knob **122** by a user of the grill **100** can nonetheless cause the first burner valve **112** to open, close, or otherwise adjust its position. In this regard, the controller **138** of FIG. **1** is configured to interpret different rotational positions of the first control knob **122** of FIG. **1** (e.g., as sensed, measured, and/or detected by the first encoder **120** of FIG. **1**) as being indicative of correlated user requests associated with different operational states (e.g., ignite, high, medium, low, or off) of the first burner **102** of FIG. **1**. For example, in response to determining that the first control knob **122** has been positioned at a relative angle of negative one hundred eighty degrees (-180°), the controller **138** may interpret the determined rotational position as a user request that the first burner **102** operate in a “medium” state. To satisfy the user request indicated by the determined rotational position of the first control knob **122**, the controller **138** may instruct, command, and/or signal the first burner valve **112** of FIG. **1** to assume a partially open (e.g., 50% open) position that facilitates a “medium” flow of gas through the first burner valve **112** and into the first burner **102**, thereby effecting the “medium” operational state of the first burner **102**.

(35) As another example, in response to determining that the first control knob **122** has been positioned at a relative angle of negative ninety degrees (-90°), the controller **138** may interpret the determined rotational position as a user request that the first burner **102** operate in a “high” state. To satisfy the user request indicated by the determined rotational position of the first control knob **122**, the controller **138** may instruct, command, and/or signal the first burner valve **112** of FIG. **1** to assume a fully open (e.g., 100% open) position that facilitates a “high” flow of gas through the first burner valve **112** and into the first burner **102**, thereby effecting the “high” operational state of the first burner **102**. As yet another example, in response to determining that the first control knob **122** has been positioned at a relative angle of zero degrees (0°), the controller **138** may interpret the determined rotational position as a user request that the first burner **102** be placed in an “off” state. To satisfy the user request indicated by the determined rotational position of the first control knob **122**, the controller **138** may instruct, command, and/or signal the first burner valve **112** of FIG. **1** to

assume a fully closed (e.g., 0% open, or 100% closed) position that prevents any flow of gas through the first burner valve **112** and into the first burner **102**, thereby effecting the “off” state of the first burner **102**.

(36) The second encoder **124** of FIG. **1** is mechanically coupled to the second control knob **126** of FIG. **1** and operatively coupled to (e.g., in electrical communication with) the controller **138** of FIG. **1**. In this regard, the second encoder **124** of FIG. **1** is implemented as a rotary encoder having a rotatable shaft to which the second control knob **126** is mechanically coupled. The rotatable shaft of the second encoder **124** can be rotated relative to a fixed portion of the second encoder **124** via user interaction with the second control knob **126** (e.g., manual rotation of the second control knob **126**). The fixed portion of the second encoder **124** includes one or more sensor(s) that is/are configured to sense, measure, and/or detect the relative angular position of the rotatable shaft and/or the relative angular position of the second control knob **126**. Data, information, and/or signals that is/are sensed, measured, and/or detected by the sensor(s) of the second encoder **124** can be transmitted directly to the controller **138** of FIG. **1**, and/or can be transmitted to and stored in the memory **144** of FIG. **1**. In some examples, the sensor(s) of the second encoder **124** is/are further configured to sense, measure, and/or detect a translational movement of the rotatable shaft relative to the fixed portion of the second encoder **124**, as may occur in response to a user of the grill **100** pushing or pressing on the second control knob **126** in a direction that is generally perpendicular to the direction(s) in which the second control knob **126** is configured to be rotated by the user.

(37) In some examples, the second encoder **124** of FIG. **1** is mounted to a control panel of the grill **100** (e.g., located along the front wall **204** of the cookbox **200** of the grill **100**) such that the second encoder **124** is located at a position on the control panel that would conventionally be occupied by a stem of a burner valve that corresponds to the second burner valve **114** of FIG. **1**. Such an example further facilitates locating the second control knob **126** of FIG. **1** at a position on or along the control panel that would conventionally be occupied by a control knob that is mechanically coupled to the stem of the burner valve that corresponds to the second burner valve **114** of FIG. **1**. While the second control knob **126** of FIG. **1** may accordingly be located at a position on or along the control panel of the grill **100** that mimics the position at which a traditional control knob is located, user actuation (e.g., manual rotation) of the second control knob **126** of FIG. **1** provides a response that differs greatly from that provided by user actuation (e.g., manual rotation) of a traditional control knob.

(38) For example, conventional multi-burner gas grills typically include a plurality of control knobs (e.g., located on or along a control panel of the grill), with each control knob being physically associated with a corresponding one of the burners of the gas grill by virtue of (1) a first mechanical connection existing between the control knob and a stem of a corresponding burner valve (e.g., such that rotation of the control knob by a user of the grill opens, closes, or otherwise adjusts the position of the burner valve), and (2) a second mechanical connection existing between the burner valve and the corresponding burner. By contrast, the grill **100** of FIG. **1** implements a “control-by-wire” architecture that eliminates the first of the aforementioned mechanical connections in favor of (1) a mechanical connection existing between the second control knob **126** of FIG. **1** and the second encoder **124** of FIG. **1**, (2) a first electrical connection existing between the second encoder **124** of FIG. **1** and the controller **138** and/or the memory **144** of FIG. **1**, and (3) a second electrical connection existing between the controller **138** of FIG. **1** and the second burner valve **114** of FIG. **1**.

(39) Although the second control knob **126** of FIG. **1** is not mechanically coupled to the second burner valve **114** of FIG. **1**, rotation of the second control knob **126** by a user of the grill **100** can nonetheless cause the second burner valve **114** to open, close, or otherwise adjust its position. In this regard, the controller **138** of FIG. **1** is configured to interpret different rotational positions of the second control knob **126** of FIG. **1** (e.g., as sensed, measured, and/or detected by the second encoder **124** of FIG. **1**) as being indicative of correlated user requests associated with different

operational states (e.g., ignite, high, medium, low, or off) of the second burner **104** of FIG. **1**. For example, in response to determining that the second control knob **126** has been positioned at a relative angle of negative one hundred eighty degrees (-180°), the controller **138** may interpret the determined rotational position as a user request that the second burner **104** operate in a “medium” state. To satisfy the user request indicated by the determined rotational position of the second control knob **126**, the controller **138** may instruct, command, and/or signal the second burner valve **114** of FIG. **1** to assume a partially open (e.g., 50% open) position that facilitates a “medium” flow of gas through the second burner valve **114** and into the second burner **104**, thereby effecting the “medium” operational state of the second burner **104**.

(40) As another example, in response to determining that the second control knob **126** has been positioned at a relative angle of negative ninety degrees (-90°), the controller **138** may interpret the determined rotational position as a user request that the second burner **104** operate in a “high” state. To satisfy the user request indicated by the determined rotational position of the second control knob **126**, the controller **138** may instruct, command, and/or signal the second burner valve **114** of FIG. **1** to assume a fully open (e.g., 100% open) position that facilitates a “high” flow of gas through the second burner valve **114** and into the second burner **104**, thereby effecting the “high” operational state of the second burner **104**. As yet another example, in response to determining that the second control knob **126** has been positioned at a relative angle of zero degrees (0°), the controller **138** may interpret the determined rotational position as a user request that the second burner **104** be placed in an “off” state. To satisfy the user request indicated by the determined rotational position of the second control knob **126**, the controller **138** may instruct, command, and/or signal the second burner valve **114** of FIG. **1** to assume a fully closed (e.g., 0% open, or 100% closed) position that prevents any flow of gas through the second burner valve **114** and into the second burner **104**, thereby effecting the “off” state of the second burner **104**.

(41) The user interface **128** of FIG. **1** includes one or more input device(s) **130** (e.g., buttons, dials, knobs, switches, touchscreens, etc.) and/or one or more output device(s) **132** (e.g., liquid crystal displays, light emitting diodes, speakers, etc.) that enable a user of the grill **100** to interact with the above-described control system of the grill **100**. In the illustrated example of FIG. **1**, the user interface **128** is operatively coupled to (e.g., in electrical communication with) the controller **138** and/or the memory **144** of the grill **100**. In some examples, the user interface **128** is mechanically coupled to (e.g., fixedly connected to) the grill **100**. For example, the user interface **128** can be mounted to the frame, to the cabinet, to one of the side tables, or even to the cookbox or a lid of the grill **100**. The user interface **128** is preferably mounted to a portion of the grill **100** that is readily accessible to a user of the grill **100**, such as a front portion of the frame, a front portion of the cabinet, a front portion of one of the side tables, a front portion of the cookbox, or a front portion of the lid of the grill **100**. In some examples, respective ones of the input device(s) **130** and/or the output device(s) **132** of the user interface **128** can be mounted to different portions of the grill **100**. For example, a first one of the input device(s) **130** can be mounted to either a rear portion or a side portion of either the frame, the cabinet, one of the side tables, the cookbox, or the lid of the grill **100**, and a second one of the input device(s) **130** can be mounted to a front portion of either the frame, the cabinet, one of the side tables, the cookbox, or the lid of the grill **100**. As another example, a first one of the input device(s) **130** can be mounted to a portion of the control panel of the grill **100** that might otherwise be occupied either by the first encoder **120** and/or the first control knob **122** of the grill **100**, or by a conventional control knob that is mechanically coupled to a conventional burner valve operatively coupled to the first burner **102** of the grill **100**, and a second one of the input device(s) **130** can be mounted to a portion of the control panel of the grill **100** that might otherwise be occupied either by the second encoder **124** and/or the second control knob **126** of the grill **100**, or by a conventional control knob that is mechanically coupled to a conventional burner valve operatively coupled to the second burner **104** of the grill **100**. The architecture and/or operations of the user interface **128** can be distributed among any number of user interfaces

respectively having any number of input device(s) **130** and/or output device(s) **132** located at and/or mounted to any portion of the grill **100**.

(42) FIG. **3** is a front view of an example implementation **300** of the user interface **128** of the grill **100** of FIG. **1**. As shown in FIG. **3**, the input device(s) **130** of the user interface **128** include an example dial **302**, an example first button **304**, an example second button **306**, and an example third button **308**, and the output device(s) **132** of the user interface **128** include an example display **310**. In the illustrated example of FIG. **3**, the dial **302** of the user interface **128** is a selection dial that can be rotated by a user of the grill **100** to adjust temperatures of the grill **100**, and/or to navigate through options presented on the display **310** of the user interface **128**. In addition to being rotatable, the dial **302** can also be pushed by a user of the grill **100** to make and/or confirm a selection of one of the options presented on the display **310**. The first button **304** of the user interface **128** is a menu button that can be pressed by a user of the grill **100** to access a main menu (e.g., a “home” menu) of selectable options, and to cause the main menu to be presented on the display **310** of the user interface **128**. The second button **306** of the user interface **128** is a cook program button that can be pressed by a user of the grill **100** to access selectable cooking programs, and to cause cooking steps associated with the selectable cooking programs to be presented on the display **310** of the user interface **128**. The third button **308** of the user interface **128** is a timer button that can be pressed by a user of the grill **100** to initiate a timer, and to cause the running time associated with the timer to be presented on the display **310** of the user interface **128**. The display **310** of the user interface **128** is a liquid crystal display configured to present textual and/or graphical information to a user of the grill **100**. In some examples, the display **310** can be implemented as a touch screen, in which case the display **310** serves not only as one of the output device(s) **132** of the user interface **128**, but also as another one of the input device(s) **130** of the user interface **128**.

(43) In some examples, one or more user input(s), selection(s), instruction(s), command(s) and/or interaction(s) can be received via the dial **302**, the first button **304**, the second button **306**, the third button **308**, or the display **310** of the user interface **128** in connection with the automated shutdown processes described herein. For example, one or more user input(s), selection(s), instruction(s), command(s) and/or interaction(s) received via the dial **302**, the first button **304**, the second button **306**, the third button **308**, or the display **310** of the user interface **128** may be indicative of a user-based shutdown request associated with the first burner **102** and/or the second burner **104** of the grill **100** and/or, more generally, associated with an automated shutdown process of the grill **100**. As another example, one or more user input(s), selection(s), instruction(s), command(s) and/or interaction(s) received via the dial **302**, the first button **304**, the second button **306**, the third button **308**, or the display **310** of the user interface **128** may be indicative of a user-based wake request associated with the control system (e.g., the user interface **128**, the controller **138**, etc.) of the grill **100** and/or, more generally, associated with an automated shutdown process of the grill **100**. As another example, one or more user input(s), selection(s), instruction(s), command(s) and/or interaction(s) received via the dial **302**, the first button **304**, the second button **306**, the third button **308**, or the display **310** of the user interface **128** may be indicative of a user-based shutdown confirmation input associated with a shutdown confirmation request and/or, more generally, associated with an automated shutdown process of the grill **100**.

(44) In some examples, one or more text-based or graphic-based notification(s) or message(s) can be presented via the display **310** of the user interface **128** in connection with the automated shutdown processes described herein. For example, one or more text-based or graphic-based notification(s) or message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that a user-based shutdown request has been received. As another example, one or more text-based or graphic-based notification(s) or message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that a user-based wake request has been received. As another example, one or more text-based or graphic-based notification(s) or

message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that the control system of the grill is in either an active (e.g., awake) state or an inactive (e.g., sleep) state. As another example, one or more text-based or graphic-based notification(s) or message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that a user-based shutdown confirmation input associated with a shutdown confirmation request has been received. As another example, one or more text-based or graphic-based notification(s) or message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that one or more burner(s) (e.g., the first burner **102** and/or the second burner **104**) of the grill **100** has/have been extinguished in connection with an automated shutdown process. As another example, one or more text-based or graphic-based notification(s) or message(s) presented via the display **310** of the user interface **128** may inform the user of the grill **100** that one or more burner(s) (e.g., the first burner **102** and/or the second burner **104**) of the grill **100** has/have not been extinguished in connection with an automated shutdown process.

(45) The network interface **134** of FIG. **1** includes one or more communication device(s) **136** (e.g., transmitter(s), receiver(s), transceiver(s), modem(s), gateway(s), wireless access point(s), etc.) to facilitate exchange of data with external machines (e.g., computing devices of any kind, including the remote device(s) **146** of FIG. **1**) by a wired or wireless communication network.

Communications transmitted and/or received via the communication device(s) **136** and/or, more generally, via the network interface **134** can be made over and/or carried by, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a wireless system, a cellular telephone system, an optical connection, etc. The network interface **134** enables a user of the grill **100** to remotely interact (e.g., via one or more of the remote device(s) **146**) with the above-described control system of the grill **100**. In the illustrated example of FIG. **1**, the network interface **134** is operatively coupled to (e.g., in electrical communication with) the controller **138** and/or the memory **144** of the grill **100**.

(46) The remote device(s) **146** of FIG. **1** can be implemented by any type(s) and/or any number(s) of mobile or stationary computing devices. In this regard, examples of such remote device(s) **146** include a smartphone, a tablet, a laptop, a desktop, a cloud server, a wearable computing device, etc. The remote device(s) **146** of FIG. **1** facilitate a remote (e.g., wired, or wireless) extension of the above-described user interface **128** of the grill **100**. In this regard, each remote device **146** includes one or more input device(s) and/or one or more output device(s) that mimic and/or enable a remotely-located version of the above-described functionality of the corresponding input device(s) **130** and/or the corresponding output device(s) **132** of the user interface **128** of the grill **100**.

Accordingly, one or more user input(s), selection(s), instruction(s), command(s) and/or interaction(s) generated via the input device(s) of the remote device(s) **146** can be received at the grill **100** (e.g., via the communication device(s) **136** of the network interface **134** of the grill **100**) in connection with the automated shutdown processes described herein. In this same regard, one or more text-based or graphic-based notification(s) or message(s) transmitted from the grill **100** (e.g., via the communication device(s) **136** of the network interface **134** of the grill **100**) can be presented via the output device(s) of the remote device(s) **146** in connection with the automated shutdown processes described herein.

(47) The controller **138** of FIG. **1** manages and/or controls the control system of the grill **100** and/or the components thereof. In the illustrated example of FIG. **1**, the controller **138** is operatively coupled to (e.g., in electrical communication with) the fuel source valve **108**, the first burner valve **112**, the second burner valve **114**, the first ignitor **116**, the second ignitor **118**, the first encoder **120**, the second encoder **124**, the user interface **128** (e.g., including the input device(s) **130** and the output device(s) **132**), the network interface **134** (e.g., including the communication device(s) **136**), and/or the memory **144** of the grill **100** of FIG. **1**. The controller **138** of FIG. **1** is also operatively coupled to (e.g., in wired or wireless electrical communication with) the remote device(s) **146** of FIG. **1** via the network interface **134** (e.g., including the communication device(s) **136**) of the grill

100 of FIG. **1**. In the illustrated example of FIG. **1**, the controller **138** includes the valve control circuitry **140** and the flame detection circuitry **142** of FIG. **1**, each of which is discussed in further detail herein. The valve control circuitry **140**, the flame detection circuitry **142**, and/or, more generally, the controller **138** of FIG. **1** can individually and/or collectively be implemented by any type(s) and/or any number(s) of semiconductor device(s) (e.g., processor(s), microprocessor(s), microcontroller(s), etc.) and/or circuit(s).

(48) In the illustrated example of FIG. **1**, the controller **138** is graphically represented as a single, discrete structure that manages and/or controls the operation(s) of various components of the control system of the grill **100**. It is to be understood, however, that in other examples, the architecture and/or operations of the controller **138** can be distributed among any number of controllers, with each separate controller having a dedicated subset of one or more operation(s) associated with the automated shutdown processes described herein. As but one example, the controller **138** of FIG. **1** can be separated into two distinct controllers, whereby a first one of the two controllers includes the valve control circuitry **140** of the controller **138**, and a second one of the two controllers includes the flame detection circuitry **142** of the controller **138**. In some examples, the grill **100** can further include separate, distinct controllers for one or more of the fuel source valve **108**, the first burner valve **112**, the second burner valve **114**, the first encoder **120**, the second encoder **124**, the user interface **128**, and/or the network interface **134** of the control system of the grill **100**.

(49) The controller **138** of FIG. **1** manages and/or controls one or more automated shutdown process(es) described herein. In some examples, the controller **138** initiates an automated shutdown process in response to the controller **138** determining that a request to shut down the grill **100** of FIG. **1** (referred to herein with reference to FIG. **1** as a “shutdown request”) has been received. For example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1**. As another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1**. As yet another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1**. In some examples, the controller **138** immediately implements one or more step(s) of an automated shutdown process in response to determining that the shutdown request has been received at the grill **100**. In other examples, the controller **138** implement(s) one or more step(s) of an automated shutdown process in a delayed manner in response to determining that the shutdown request has been received at the grill **100**, with the associated implementation delay(s) being based on a shutdown schedule and/or a shutdown time indicated by the shutdown request.

(50) In some examples, the controller **138** of FIG. **1** determines whether a request to wake the control system of the grill **100** of FIG. **1** (referred to herein with reference to FIG. **1** as a “wake request”) has been received. For example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1**. As another example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a

pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1**. As yet another example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1**.

(51) In some examples, the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes the control system of the grill **100** of FIG. **1** to wake in response to the controller **138** determining that a wake request has been received. For example, the controller **138** may instruct, command, signal, and/or otherwise cause the control system of the grill **100** to transition from an inactive (e.g., sleep) state to an active (e.g., awake) state in response to the controller **138** determining that a wake request has been received. In some examples, the controller **138** of FIG. **1** recognizes, responds to, and/or otherwise acts upon a shutdown request that has been received at the grill **100** of FIG. **1** only when the controller **138** first determines that the shutdown request was received at the grill **100** at a time when the control system of the grill **100** is/was awake. In instances where the controller **138** determines that the shutdown request was received at the grill **100** at a time when the control system of the grill **100** is/was not awake, the controller **138** declines to recognize, respond to, and/or otherwise act on the shutdown request, and instead waits for a wake request to be received at the grill **100**. In still other examples, the controller **138** can instead recognize, respond to, and/or otherwise act on a shutdown request that has been received at the grill **100** of FIG. **1** even if the controller **138** determines that the startup request was received at the grill **100** at a time when the control system of the grill **100** is/was in an inactive (e.g., sleep) state.

(52) In some examples, the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes the user interface **128** of the grill **100** of FIG. **1** to present a request for a user-based shutdown confirmation input (referred to herein with reference to FIG. **1** as a “shutdown confirmation request”). In some examples, the controller **138** of FIG. **1** additionally or alternatively causes the network interface **134** of the grill **100** to transmit the shutdown confirmation request to one or more of the remote device(s) **146** of FIG. **1**. In some examples, the shutdown confirmation request transmitted via the network interface **134** may be the same as, or may mimic the shutdown confirmation request presented via the user interface **128**. In some examples, the controller **138** initiates a timer in association with the shutdown confirmation request, whereby the timer provides and/or imposes a time frame, a time period, and/or a time limit by which a response to the shutdown confirmation request must be made and/or received at the grill **100**.

(53) In some examples, the controller **138** of FIG. **1** determines whether a shutdown confirmation input has been received at the grill **100** of FIG. **1** before the timer associated with the shutdown confirmation request times out. For example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request times out based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1** occurring before the timer associated with the shutdown confirmation request times out. As another example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request times out based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1** before the timer associated with the shutdown confirmation request times out. As yet another example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request times out based on a user input,

a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1** before the timer associated with the shutdown confirmation request times out. In some examples, the controller **138** of FIG. **1** recognizes, responds to, and/or otherwise acts upon a shutdown request that has been received at the grill **100** of FIG. **1** only when the controller **138** first determines that the shutdown confirmation input was received at the grill **100** of FIG. **1** before the timer associated with the shutdown confirmation request times out. In instances where the controller **138** determines that the shutdown confirmation input was not received at the grill **100** of FIG. **1** before the timer associated with the shutdown confirmation request times out, the controller **138** declines to recognize, respond to, and/or otherwise act on the shutdown request, and instead waits for another shutdown request to be received at the grill **100**.

(54) The valve control circuitry **140** of the controller **138** of FIG. **1** manages and/or controls one or more operation(s) of the first burner valve **112** and/or the second burner valve **114** of the grill **100** of FIG. **1** in association with one or more automated shutdown process(es) described herein. In some examples, the valve control circuitry **140** of the controller **138** instructs, commands, signals, and/or otherwise causes multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to close. For example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to concurrently close (e.g., the first burner valve **112** and the second burner valve **114** closing at substantially the same time). As another example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to sequentially close (e.g., the first burner valve **112** closing at a first time, followed by the second burner valve **114** closing at a second time subsequent to the first time). In some examples, the valve control circuitry **140** of the controller **138** of FIG. **1** also manages and/or controls one or more operation(s) of the fuel source valve **108** of the grill **100** of FIG. **1** in association with one or more automated shutdown process(es) described herein. For example, the valve control circuitry **140** of the controller **138** can instruct, command, signal, and/or otherwise cause the fuel source valve **108** to close.

(55) The flame detection circuitry **142** of the controller **138** of FIG. **1** senses and/or detects whether the first burner **102** and/or the second burner **104** of the grill **100** of FIG. **1** is/are extinguished (e.g., not ignited). In some examples, the controller **138** relies on the flame detection circuitry **142** in connection with the controller **138** determining whether one or more operation(s) associated with an automated shutdown process was/were successfully performed (e.g., whether the first burner **102** and/or the second burner **104** was/were successfully extinguished). For example, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and the second burner **104** of the grill **100** have been extinguished in connection with an automated shutdown process, the controller **138** can accordingly determine that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were successfully performed. Conversely, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and/or the second burner **104** of the grill **100** has/have not been extinguished in connection with the automated shutdown process, the controller **138** can accordingly determine that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were not successfully performed.

(56) The flame detection circuitry **142** of the controller **138** of FIG. **1** can accordingly be utilized as a diagnostic tool with regard to determining and/or addressing the success and/or failure of any

automated shutdown process described herein. In some examples, an indication from the flame detection circuitry **142** that one or more operation(s) associated with an automated shutdown process has/have failed may cause the controller **138** to re-instruct, re-command, and/or re-signal one or more component(s) of the control system of the grill **100** to repeat one or more previously-instructed, previously-commanded, and/or previously-signaled operation(s) associated with the automated shutdown process. The flame detection circuitry **142** of the controller of FIG. **1** can be structured, configured, and/or implemented as described in U.S. patent application Ser. No. 17/144,038, filed on Jan. 7, 2021. The entirety of U.S. patent application Ser. No. 17/144,038 is hereby incorporated by reference herein.

(57) The memory **144** of FIG. **1** can be implemented by any type(s) and/or any number(s) of storage device(s) such as a storage drive, a flash memory, a read-only memory (ROM), a random-access memory (RAM), a cache and/or any other physical storage medium in which information is stored for any duration (e.g., for extended time periods, permanently, brief instances, for temporarily buffering, and/or for caching of the information). The information stored in the memory **144** of FIG. **1** can be stored in any file and/or data structure format, organization scheme, and/or arrangement.

(58) The memory **144** stores data sensed, measured, detected, generated, input, output, transmitted, and/or received by, to, and/or from the fuel source valve **108**, the first burner valve **112**, the second burner valve **114**, the first ignitor **116**, the second ignitor **118**, the first encoder **120**, the second encoder **124**, the user interface **128** (e.g., including the input device(s) **130** and the output device(s) **132**), the network interface **134** (e.g., including the communication device(s) **136**), the controller **138** (e.g., including the valve control circuitry **140** and the flame detection circuitry **142**), the remote device(s) **146**, and/or, more generally, the control system of the grill **100** of FIG. **1**. The memory **144** also stores instructions (e.g., machine-readable instructions) and associated data corresponding to the processes, protocols, sequences and/or methods described below in connection with FIGS. **4-7**. The memory **144** of FIG. **1** is accessible to one or more of the fuel source valve **108**, the first burner valve **112**, the second burner valve **114**, the first encoder **120**, the second encoder **124**, the user interface **128** (e.g., including the input device(s) **130** and the output device(s) **132**), the network interface **134** (e.g., including the communication device(s) **136**), the controller **138** (e.g., including the valve control circuitry **140** and the flame detection circuitry **142**), the remote device(s) **146**, and/or, more generally, the control system of the grill **100** of FIG. **1**.

(59) While an example manner of implementing the control system of the grill **100** is illustrated in FIG. **1**, one or more of the elements, processes, and/or devices illustrated in FIG. **1** may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the example fuel source valve **108**, the example first burner valve **112**, the example second burner valve **114**, the example first ignitor **116**, the example second ignitor **118**, the example first encoder **120**, the example second encoder **124**, the example user interface **128** (e.g., including the example input device(s) **130** and the example output device(s) **132**), the example network interface **134** (e.g., including the example communication device(s) **136**), the example controller **138** (e.g., including the example valve control circuitry **140** and the example flame detection circuitry **142**), the example memory **144**, and/or, more generally, the control system of the grill **100** of FIG. **1**, may be implemented by hardware, software, firmware, and/or any combination of hardware, software, and/or firmware. Thus, for example, any of the example fuel source valve **108**, the example first burner valve **112**, the example second burner valve **114**, the example first ignitor **116**, the example second ignitor **118**, the example first encoder **120**, the example second encoder **124**, the example user interface **128** (e.g., including the example input device(s) **130** and the example output device(s) **132**), the example network interface **134** (e.g., including the example communication device(s) **136**), the example controller **138** (e.g., including the example valve control circuitry **140** and the example flame detection circuitry **142**), the example memory **144**, and/or, more generally, the control system of the grill **100** of FIG. **1**, could be implemented by processor circuitry, analog

circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as Field Programmable Gate Arrays (FPGAs). When reading any of the apparatus or system claims of this patent to cover a purely software and/or firmware implementation, at least one of the example fuel source valve **108**, the example first burner valve **112**, the example second burner valve **114**, the example first ignitor **116**, the example second ignitor **118**, the example first encoder **120**, the example second encoder **124**, the example user interface **128** (e.g., including the example input device(s) **130** and the example output device(s) **132**), the example network interface **134** (e.g., including the example communication device(s) **136**), the example controller **138** (e.g., including the example valve control circuitry **140** and the example flame detection circuitry **142**), and/or the example memory **144** is/are hereby expressly defined to include a non-transitory computer readable storage device or storage disk such as a memory, a digital versatile disk (DVD), a compact disk (CD), a Blu-ray disk, etc., including the software and/or firmware. Further still, the example control system of the grill of FIG. **1** may include one or more elements, processes, and/or devices in addition to, or instead of, those illustrated in FIG. **1**, and/or may include more than one of any or all of the illustrated elements, processes, and devices.

(60) Flowcharts representative of example hardware logic circuitry, machine-readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the control system of the grill **100** of FIG. **1** are shown in FIGS. **4-7**. The machine-readable instructions may be one or more executable programs or portion(s) of an executable program for execution by processor circuitry, such as the processor circuitry **802** shown in the example processor platform **800** discussed below in connection with FIG. **8** and/or the example processor circuitry discussed below in connection with FIGS. **9** and/or **10**. The program may be embodied in software stored on one or more non-transitory computer readable storage media such as a CD, a floppy disk, a hard disk drive (HDD), a DVD, a Blu-ray disk, a volatile memory (e.g., Random Access Memory (RAM) of any type, etc.), or a non-volatile memory (e.g., FLASH memory, an HDD, etc.) associated with processor circuitry located in one or more hardware devices, but the entire program and/or parts thereof could alternatively be executed by one or more hardware devices other than the processor circuitry and/or embodied in firmware or dedicated hardware. The machine-readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a user) or an intermediate client hardware device (e.g., a radio access network (RAN) gateway that may facilitate communication between a server and an endpoint client hardware device). Similarly, the non-transitory computer readable storage media may include one or more mediums located in one or more hardware devices. Further, although example programs are described with reference to the flowcharts illustrated in FIGS. **4-7**, many other methods of implementing the example control system of the grill **100** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally, or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The processor circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core central processor unit (CPU)), a multi-core processor (e.g., a multi-core CPU), etc.) in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, a CPU and/or a FPGA located in the same

package (e.g., the same integrated circuit (IC) package or in two or more separate housings, etc.).

(61) The machine-readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine-readable instructions as described herein may be stored as data or a data structure (e.g., as portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine-executable instructions. For example, the machine-readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine-readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or any other machine. For example, the machine-readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of machine-executable instructions that implement one or more operations that may together form a program such as that described herein.

(62) In another example, the machine-readable instructions may be stored in a state in which they may be read by processor circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine-readable instructions on a particular computing device or any other device. In another example, the machine-readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine-readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine-readable media, as used herein, may include machine-readable instructions and/or program(s) regardless of the particular format or state of the machine-readable instructions and/or program(s) when stored or otherwise at rest or in transit.

(63) The machine-readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine-readable instructions may be represented using any of the following languages: C, C++, Java, C#, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

(64) As mentioned above, the example operations of FIGS. 4-7 may be implemented using executable instructions (e.g., computer and/or machine-readable instructions) stored on one or more non-transitory computer and/or machine-readable media such as optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory (ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms “non-transitory computer-readable medium” and “non-transitory computer-readable storage medium” are expressly defined to include any type of computer-readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

(65) The terms “including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A,

B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

(66) As used herein, singular references (e.g., “a,” “an,” “first,” “second,” etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or method actions may be implemented by, for example, the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

(67) FIG. 4 is a flowchart representative of example machine-readable instructions and/or example operations **400** that may be executed and/or instantiated by processor circuitry to implement an automated shutdown process via the grill **100** of FIG. 1. The machine-readable instructions and/or operations **400** of FIG. 4 begin at block **402** when the controller **138** of FIG. 1 determines whether a request to shut down the grill **100** of FIG. 1 (referred to herein with reference to FIG. 4 as a “shutdown request”) has been received. For example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. 1. As another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. 1, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. 1. As yet another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. 1, as received and/or detected via the network interface **134** of FIG. 1. If the controller **138** determines at block **402** that a shutdown request has not been received, control of the machine-readable instructions and/or operations **400** of FIG. 4 remains at block **402**. If the controller **138** instead determines at block **402** that a shutdown request has been received, control of the machine-readable instructions and/or operations **400** of FIG. 4 proceeds to block **404**.

(68) At block **404**, the valve control circuitry **140** of the controller **138** of FIG. 1 instructs, commands, signals, and/or otherwise causes multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. 1 to close. For example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. 1 to concurrently close (e.g., the first

burner valve **112** and the second burner valve **114** closing at substantially the same time). As another example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to sequentially close (e.g., the first burner valve **112** closing at a first time, followed by the second burner valve **114** closing at a second time subsequent to the first time). Following block **404**, the machine-readable instructions and/or operations **400** of FIG. **4** end.

(69) In the above-described example, the machine-readable instructions and/or operations **400** of FIG. **4** end at block **404**. In other examples, the machine-readable instructions and/or operations **400** of FIG. **4** can instead include one or more additional blocks (e.g., following block **404**) in which the flame detection circuitry **142** of the controller **138** of FIG. **1** determines whether multiple ones (e.g., all) of the burners (e.g., the first burner **102** and the second burner **104**) of the grill **100** of FIG. **1** are extinguished (e.g., not ignited). If the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and the second burner **104** of the grill **100** have been extinguished in connection with the automated shutdown process of FIG. **4**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were successfully performed. Conversely, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and/or the second burner **104** of the grill **100** has/have not been extinguished in connection with the automated shutdown process of FIG. **4**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were not successfully performed. The flame detection circuitry **142** of the controller **138** of FIG. **1** can accordingly be utilized as a diagnostic tool with regard to determining and/or addressing the success and/or failure of the automated shutdown process of FIG. **4**. In some examples, an indication from the flame detection circuitry **142** that one or more operation(s) associated with the automated shutdown process of FIG. **4** has/have failed may cause the controller **138** to re-instruct, re-command, and/or re-signal one or more component(s) of the control system of the grill **100** to repeat one or more previously-instructed, previously-commanded, and/or previously-signaled operation(s) associated with the automated shutdown process of FIG. **4**.

(70) FIG. **5** is a flowchart representative of example machine-readable instructions and/or example operations **500** that may be executed and/or instantiated by processor circuitry to implement another automated shutdown process via the grill **100** of FIG. **1**. The machine-readable instructions and/or operations **500** of FIG. **5** begin at block **502** when the controller **138** of FIG. **1** determines whether a request to wake the control system of the grill **100** of FIG. **1** (referred to herein with reference to FIG. **5** as a “wake request”) has been received. For example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1**. As another example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1**. As yet another example, the controller **138** may determine that a wake request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1**. If the controller **138** determines at block **502** that a wake request has not been received, control of the machine-readable instructions and/or operations **500** of FIG. **5** remains at

block **502**. If the controller **138** instead determines at block **502** that a wake request has been received, control of the machine-readable instructions and/or operations **500** of FIG. 5 proceeds to block **504**.

(71) At block **504**, the controller **138** of FIG. 1 instructs, commands, signals, and/or otherwise causes the control system of the grill **100** of FIG. 1 to wake. For example, the controller **138** of FIG. 1 may instruct, command, signal, and/or otherwise cause the control system of the grill **100** of FIG. 1 to transition from an inactive (e.g., sleep) state to an active (e.g., awake) state. Following block **504**, control of the example machine-readable instructions and/or operations **500** of FIG. 5 proceeds to block **506**.

(72) At block **506**, the controller **138** of FIG. 1 determines whether a request to shut down the grill **100** of FIG. 1 (referred to herein with reference to FIG. 5 as a “shutdown request”) has been received while the control system of the grill **100** of FIG. 1 is awake. For example, the controller **138** may determine that a shutdown request has been received while the control system of the grill **100** is awake based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. 1 occurring while the control system of the grill **100** is awake. As another example, the controller **138** may determine that a shutdown request has been received while the control system of the grill **100** is awake based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. 1, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. 1 while the control system of the grill **100** is awake. As yet another example, the controller **138** may determine that a shutdown request has been received while the control system of the grill **100** is awake based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. 1, as received and/or detected via the network interface **134** of FIG. 1 while the control system of the grill **100** is awake. If the controller **138** determines at block **506** that a shutdown request has not been received while the control system of the grill **100** is awake, control of the machine-readable instructions and/or operations **500** of FIG. 5 returns to block **502**. If the controller **138** instead determines at block **506** that a shutdown request has been received while the control system of the grill **100** is awake, control of the machine-readable instructions and/or operations **500** of FIG. 5 proceeds to block **508**.

(73) At block **508**, the valve control circuitry **140** of the controller **138** of FIG. 1 instructs, commands, signals, and/or otherwise causes multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. 1 to close. For example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. 1 to concurrently close (e.g., the first burner valve **112** and the second burner valve **114** closing at substantially the same time). As another example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. 1 to sequentially close (e.g., the first burner valve **112** closing at a first time, followed by the second burner valve **114** closing at a second time subsequent to the first time). Following block **508**, the machine-readable instructions and/or operations **500** of FIG. 5 end.

(74) In the above-described example, the machine-readable instructions and/or operations **500** of FIG. 5 end at block **508**. In other examples, the machine-readable instructions and/or operations **500** of FIG. 5 can instead include one or more additional blocks (e.g., following block **508**) in which the flame detection circuitry **142** of the controller **138** of FIG. 1 determines whether multiple

ones (e.g., all) of the burners (e.g., the first burner **102** and the second burner **104**) of the grill **100** of FIG. **1** are extinguished (e.g., not ignited). If the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and the second burner **104** of the grill **100** have been extinguished in connection with the automated shutdown process of FIG. **5**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were successfully performed. Conversely, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and/or the second burner **104** of the grill **100** has/have not been extinguished in connection with the automated shutdown process of FIG. **5**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were not successfully performed. The flame detection circuitry **142** of the controller **138** of FIG. **1** can accordingly be utilized as a diagnostic tool with regard to determining and/or addressing the success and/or failure of the automated shutdown process of FIG. **5**. In some examples, an indication from the flame detection circuitry **142** that one or more operation(s) associated with the automated shutdown process of FIG. **5** has/have failed may cause the controller **138** to re-instruct, re-command, and/or re-signal one or more component(s) of the control system of the grill **100** to repeat one or more previously-instructed, previously-commanded, and/or previously-signaled operation(s) associated with the automated shutdown process of FIG. **5**.

(75) FIG. **6** is a flowchart representative of example machine-readable instructions and/or example operations **600** that may be executed and/or instantiated by processor circuitry to implement another automated shutdown process via the grill **100** of FIG. **1**. The machine-readable instructions and/or operations **600** of FIG. **6** begin at block **602** when the controller **138** of FIG. **1** determines whether a request to shut down the grill **100** of FIG. **1** (referred to herein with reference to FIG. **6** as a “shutdown request”) has been received. For example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1**. As another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1**. As yet another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1**. If the controller **138** determines at block **602** that a shutdown request has not been received, control of the machine-readable instructions and/or operations **600** of FIG. **6** remains at block **602**. If the controller **138** instead determines at block **602** that a shutdown request has been received, control of the machine-readable instructions and/or operations **600** of FIG. **6** proceeds to block **604**.

(76) At block **604**, the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes the user interface **128** of the grill **100** of FIG. **1** to present a request for a user-based shutdown confirmation input (referred to herein with reference to FIG. **6** as a “shutdown confirmation request”). In some examples, the controller **138** of FIG. **1** additionally or alternatively causes the network interface **134** of the grill **100** to transmit the shutdown confirmation request to one or more of the remote device(s) **146** of FIG. **1**. In some examples, the shutdown confirmation request transmitted via the network interface **134** may be the same as, or may mimic the shutdown confirmation request presented via the user interface **128**. In some examples, the controller **138** initiates a timer at block **604** in association with the shutdown confirmation request, whereby the

timer provides and/or imposes a time frame, a time period, and/or a time limit by which a response to the shutdown confirmation request must be made and/or received. Following block **604**, control of the machine-readable instructions and/or operations **600** of FIG. **6** proceeds to block **606**.

(77) At block **606**, the controller **138** of FIG. **1** determines whether a shutdown confirmation input has been received at the grill **100** before the timer associated with the shutdown confirmation request of block **604** times out. For example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request of block **604** times out based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1** occurring before the timer associated with the shutdown confirmation request of block **604** times out. As another example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request of block **604** times out based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1** before the timer associated with the shutdown confirmation request of block **604** times out. As yet another example, the controller **138** may determine that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request of block **604** times out based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1** before the timer associated with the shutdown confirmation request of block **604** times out. If the controller **138** determines at block **606** that a shutdown confirmation input has not been received at the grill **100** before the timer associated with the shutdown confirmation request of block **604** times out, control of the machine-readable instructions and/or operations **600** of FIG. **6** returns to block **602**. If the controller **138** instead determines at block **606** that a shutdown confirmation input has been received before the timer associated with the shutdown confirmation request of block **604** times out, control of the machine-readable instructions and/or operations **600** of FIG. **6** proceeds to block **608**.

(78) At block **608**, the valve control circuitry **140** of the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to close. For example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to concurrently close (e.g., the first burner valve **112** and the second burner valve **114** closing at substantially the same time). As another example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to sequentially close (e.g., the first burner valve **112** closing at a first time, followed by the second burner valve **114** closing at a second time subsequent to the first time). Following block **608**, the machine-readable instructions and/or operations **600** of FIG. **6** end.

(79) In the above-described example, the machine-readable instructions and/or operations **600** of FIG. **6** end at block **608**. In other examples, the machine-readable instructions and/or operations **600** of FIG. **6** can instead include one or more additional blocks (e.g., following block **608**) in which the flame detection circuitry **142** of the controller **138** of FIG. **1** determines whether multiple ones (e.g., all) of the burners (e.g., the first burner **102** and the second burner **104**) of the grill **100** of FIG. **1** are extinguished (e.g., not ignited). If the flame detection circuitry **142** of the controller

138 indicates that the first burner **102** and the second burner **104** of the grill **100** have been extinguished in connection with the automated shutdown process of FIG. **6**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were successfully performed. Conversely, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and/or the second burner **104** of the grill **100** has/have not been extinguished in connection with the automated shutdown process of FIG. **6**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were not successfully performed. The flame detection circuitry **142** of the controller **138** of FIG. **1** can accordingly be utilized as a diagnostic tool with regard to determining and/or addressing the success and/or failure of the automated shutdown process of FIG. **6**. In some examples, an indication from the flame detection circuitry **142** that one or more operation(s) associated with the automated shutdown process of FIG. **6** has/have failed may cause the controller **138** to re-instruct, re-command, and/or re-signal one or more component(s) of the control system of the grill **100** to repeat one or more previously-instructed, previously-commanded, and/or previously-signaled operation(s) associated with the automated shutdown process of FIG. **6**.

(80) FIG. **7** is a flowchart representative of example machine-readable instructions and/or example operations **700** that may be executed and/or instantiated by processor circuitry to implement another automated shutdown process via the grill **100** of FIG. **1**. The machine-readable instructions and/or operations **700** of FIG. **7** begin at block **702** when the controller **138** of FIG. **1** determines whether a request to shut down the grill **100** of FIG. **1** (referred to herein with reference to FIG. **7** as a “shutdown request”) has been received. For example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more of the input device(s) **130** (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of the user interface **128** of FIG. **1**. As another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, etc.) of, to, and/or with the first control knob **122** and/or the second control knob **126** of FIG. **1**, as sensed, measured, and or detected via the first encoder **120** and/or the second encoder **124** of FIG. **1**. As yet another example, the controller **138** may determine that a shutdown request has been received based on a user input, a user selection, and/or a user interaction (e.g., a press, a push, a pull, a rotation, a click, a flip, a swipe, a touch, etc.) of, to, and/or with one or more input device(s) (e.g., a button, a dial, a knob, a switch, a touchscreen, etc.) of one of the remote device(s) **146** of FIG. **1**, as received and/or detected via the network interface **134** of FIG. **1**. If the controller **138** determines at block **702** that a shutdown request has not been received, control of the machine-readable instructions and/or operations **700** of FIG. **7** remains at block **702**. If the controller **138** instead determines at block **702** that a shutdown request has been received, control of the machine-readable instructions and/or operations **700** of FIG. **7** proceeds to block **704**.

(81) At block **704**, the valve control circuitry **140** of the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to close. For example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to concurrently close (e.g., the first burner valve **112** and the second burner valve **114** closing at substantially the same time). As another example, the valve control circuitry **140** of the controller **138** may instruct, command, signal, and/or otherwise cause multiple ones (e.g., all) of the burner valves (e.g., the first burner valve **112** and the second burner valve **114**) of the grill **100** of FIG. **1** to sequentially close (e.g., the first burner valve **112** closing at a first time, followed by the second burner valve **114** closing at a

second time subsequent to the first time). Following block **704**, control of the machine-readable instructions and/or operations **700** of FIG. **7** proceeds to block **706**.

(82) At block **706**, the valve control circuitry **140** of the controller **138** of FIG. **1** instructs, commands, signals, and/or otherwise causes the fuel source valve **108** of the grill **100** of FIG. **1** to close. Following block **704**, the machine-readable instructions and/or operations **700** of FIG. **7** end.

(83) In the above-described example, the machine-readable instructions and/or operations **700** of FIG. **7** end at block **706**. In other examples, the machine-readable instructions and/or operations **700** of FIG. **7** can instead include one or more additional blocks (e.g., following block **706**) in which the flame detection circuitry **142** of the controller **138** of FIG. **1** determines whether multiple ones (e.g., all) of the burners (e.g., the first burner **102** and the second burner **104**) of the grill **100** of FIG. **1** are extinguished (e.g., not ignited). If the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and the second burner **104** of the grill **100** have been extinguished in connection with the automated shutdown process of FIG. **7**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were successfully performed. Conversely, if the flame detection circuitry **142** of the controller **138** indicates that the first burner **102** and/or the second burner **104** of the grill **100** has/have not been extinguished in connection with the automated shutdown process of FIG. **7**, the controller **138** determines that one or more valve control operation(s) (e.g., controlled by the valve control circuitry **140** of the controller **138**) associated with the automated shutdown process were not successfully performed. The flame detection circuitry **142** of the controller **138** of FIG. **1** can accordingly be utilized as a diagnostic tool with regard to determining and/or addressing the success and/or failure of the automated shutdown process of FIG. **7**. In some examples, an indication from the flame detection circuitry **142** that one or more operation(s) associated with the automated shutdown process of FIG. **7** has/have failed may cause the controller **138** to re-instruct, re-command, and/or re-signal one or more component(s) of the control system of the grill **100** to repeat one or more previously-instructed, previously-commanded, and/or previously-signaled operation(s) associated with the automated shutdown process of FIG. **7**.

(84) FIG. **8** is a block diagram of an example processor platform **800** including processor circuitry structured to execute and/or instantiate the machine-readable instructions and/or operations of FIGS. **4-7** to implement the grill **100** of FIG. **1**. The processor platform **800** of the illustrated example includes processor circuitry **802**. The processor circuitry **802** of the illustrated example is hardware. For example, the processor circuitry **802** can be implemented by one or more integrated circuit(s), logic circuit(s), FPGA(s), microprocessor(s), CPU(s), GPU(s), DSP(s), and/or microcontroller(s) from any desired family or manufacturer. The processor circuitry **802** may be implemented by one or more semiconductor based (e.g., silicon based) device(s). In this example, the processor circuitry **802** implements the controller **138** of FIG. **1**, including the valve control circuitry **140** and the flame detection circuitry **142** of the controller **138**.

(85) The processor circuitry **802** of the illustrated example includes a local memory **804** (e.g., a cache, registers, etc.). The processor circuitry **802** is in electrical communication with one or more valve(s) **806** via a bus **808**. In this example, the valve(s) **806** include the fuel source valve **108**, the first burner valve **112**, and the second burner valve **114** of FIG. **1**. The processor circuitry **802** is also in electrical communication with one or more ignitor(s) **810** via the bus **808**. In this example, the ignitor(s) **810** include the first ignitor **116** and the second ignitor **118** of FIG. **1**. The processor circuitry **802** is also in electrical communication with one or more encoder(s) **812** via the bus **808**. In this example, the encoder(s) **812** include the first encoder **120** and the second encoder **124** of FIG. **1**.

(86) The processor circuitry **802** is also in electrical communication with a main memory via the bus **808**, with the main memory including a volatile memory **814** and a non-volatile memory **816**. The volatile memory **814** may be implemented by Synchronous Dynamic Random Access Memory

(SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory **816** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **814**, **816** of the illustrated example is controlled by a memory controller.

(87) The processor platform **800** of the illustrated example also includes one or more mass storage device(s) **818** to store software and/or data. Examples of such mass storage device(s) **818** include magnetic storage devices, optical storage devices, floppy disk drives, HDDs, CDs, Blu-ray disk drives, redundant array of independent disks (RAID) systems, solid state storage devices such as flash memory devices, and DVD drives. In the illustrated example of FIG. **8**, one or more of the volatile memory **814**, the non-volatile memory **816**, and/or the mass storage device(s) **818** implement(s) the memory **144** of FIG. **1**.

(88) The processor platform **800** of the illustrated example also includes user interface circuitry **820**. The user interface circuitry **820** may be implemented by hardware in accordance with any type of interface standard, such as an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a PCI interface, and/or a PCIe interface. In the illustrated example, one or more input device(s) **130** are connected to the user interface circuitry **820**. The input device(s) **130** permit(s) a user to enter data and/or commands into the processor circuitry **802**. The input device(s) **130** can be implemented by, for example, one or more button(s), dial(s), knob(s), switch(es), touchscreen(s), audio sensor(s), microphone(s), image sensor(s), and/or camera(s). One or more output device(s) **132** are also connected to the user interface circuitry **820** of the illustrated example. The output device(s) **132** can be implemented, for example, by one or more display device(s) (e.g., light emitting diode(s) (LED(s)), organic light emitting diode(s) (OLED(s)), liquid crystal display(s) (LCD(s)), cathode ray tube (CRT) display(s), in-place switching (IPS) display(s), touchscreen(s), etc.), tactile output device(s), and/or speaker(s). The user interface circuitry **820** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip, and/or graphics processor circuitry such as a GPU. In the illustrated example of FIG. **8**, the user interface circuitry **820**, the input device(s) **130**, and the output device(s) **132** collectively implement the user interface **128** of FIG. **1**.

(89) The processor platform **800** of the illustrated example also includes network interface circuitry **822**. The network interface circuitry **822** includes one or more communication device(s) (e.g., transmitter(s), receiver(s), transceiver(s), modem(s), gateway(s), wireless access point(s), etc.) to facilitate exchange of data with external machines (e.g., computing devices of any kind, including the remote device(s) **146** of FIG. **1**) by a network **824**. The communication can be by, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a wireless system, a cellular telephone system, an optical connection, etc. In the illustrated example of FIG. **8**, the network interface circuitry **822** implements the network interface **134** (e.g., including the communication device(s) **136**) of FIG. **1**.

(90) Coded instructions **826** including the above-described machine-readable instructions and/or operations of FIGS. **4-7** may be stored in the local memory **804**, in the volatile memory **814**, in the non-volatile memory **816**, on the mass storage device(s) **818**, and/or on a removable non-transitory computer-readable storage medium such as a flash memory stick, a dongle, a CD, or a DVD.

(91) FIG. **9** is a block diagram of an example implementation of the processor circuitry **802** of FIG. **8**. In this example, the processor circuitry **802** of FIG. **8** is implemented by a microprocessor **900**. For example, the microprocessor **900** may implement multi-core hardware circuitry such as a CPU, a DSP, a GPU, an XPU, etc. Although it may include any number of example cores **902** (e.g., **1** core), the microprocessor **900** of this example is a multi-core semiconductor device including **N** cores. The cores **902** of the microprocessor **900** may operate independently or may cooperate to execute machine-readable instructions. For example, machine code corresponding to a firmware program, an embedded software program, or a software program may be executed by one of the cores **902** or may be executed by multiple ones of the cores **902** at the same or different times. In

some examples, the machine code corresponding to the firmware program, the embedded software program, or the software program is split into threads and executed in parallel by two or more of the cores **902**. The software program may correspond to a portion or all of the machine-readable instructions and/or operations represented by the flowcharts of FIGS. 4-7.

(92) The cores **902** may communicate by an example bus **904**. In some examples, the bus **904** may implement a communication bus to effectuate communication associated with one(s) of the cores **902**. For example, the bus **904** may implement at least one of an Inter-Integrated Circuit (I2C) bus, a Serial Peripheral Interface (SPI) bus, a PCI bus, or a PCIe bus. Additionally, or alternatively, the bus **904** may implement any other type of computing or electrical bus. The cores **902** may obtain data, instructions, and/or signals from one or more external devices by example interface circuitry **906**. The cores **902** may output data, instructions, and/or signals to the one or more external devices by the interface circuitry **906**. Although the cores **902** of this example include example local memory **920** (e.g., Level 1 (L1) cache that may be split into an L1 data cache and an L1 instruction cache), the microprocessor **900** also includes example shared memory **910** that may be shared by the cores (e.g., Level 2 (L2_cache)) for high-speed access to data and/or instructions. Data and/or instructions may be transferred (e.g., shared) by writing to and/or reading from the shared memory **910**. The local memory **920** of each of the cores **902** and the shared memory **910** may be part of a hierarchy of storage devices including multiple levels of cache memory and the main memory (e.g., the main memory **814, 816** of FIG. 8). Typically, higher levels of memory in the hierarchy exhibit lower access time and have smaller storage capacity than lower levels of memory. Changes in the various levels of the cache hierarchy are managed (e.g., coordinated) by a cache coherency policy.

(93) Each core **902** may be referred to as a CPU, DSP, GPU, etc., or any other type of hardware circuitry. Each core **902** includes control unit circuitry **914**, arithmetic and logic (AL) circuitry (sometimes referred to as an ALU) **916**, a plurality of registers **918**, the L1 cache **920**, and an example bus **922**. Other structures may be present. For example, each core **902** may include vector unit circuitry, single instruction multiple data (SIMD) unit circuitry, load/store unit (LSU) circuitry, branch/jump unit circuitry, floating-point unit (FPU) circuitry, etc. The control unit circuitry **914** includes semiconductor-based circuits structured to control (e.g., coordinate) data movement within the corresponding core **902**. The AL circuitry **916** includes semiconductor-based circuits structured to perform one or more mathematic and/or logic operations on the data within the corresponding core **902**. The AL circuitry **916** of some examples performs integer based operations. In other examples, the AL circuitry **916** also performs floating point operations. In yet other examples, the AL circuitry **916** may include first AL circuitry that performs integer based operations and second AL circuitry that performs floating point operations. In some examples, the AL circuitry **916** may be referred to as an Arithmetic Logic Unit (ALU). The registers **918** are semiconductor-based structures to store data and/or instructions such as results of one or more of the operations performed by the AL circuitry **916** of the corresponding core **902**. For example, the registers **918** may include vector register(s), SIMD register(s), general purpose register(s), flag register(s), segment register(s), machine specific register(s), instruction pointer register(s), control register(s), debug register(s), memory management register(s), machine check register(s), etc. The registers **918** may be arranged in a bank as shown in FIG. 9. Alternatively, the registers **918** may be organized in any other arrangement, format, or structure including distributed throughout the core **902** to shorten access time. The bus **922** may implement at least one of an I2C bus, a SPI bus, a PCI bus, or a PCIe bus.

(94) Each core **902** and/or, more generally, the microprocessor **900** may include additional and/or alternate structures to those shown and described above. For example, one or more clock circuits, one or more power supplies, one or more power gates, one or more cache home agents (CHAs), one or more converged/common mesh stops (CMSs), one or more shifters (e.g., barrel shifter(s)), and/or other circuitry may be present. The microprocessor **900** is a semiconductor device fabricated to include many transistors interconnected to implement the structures described above in one or

more integrated circuits (ICs) contained in one or more packages. The processor circuitry may include and/or cooperate with one or more accelerators. In some examples, accelerators are implemented by logic circuitry to perform certain tasks more quickly and/or efficiently than can be done by a general purpose processor. Examples of accelerators include ASICs and FPGAs such as those discussed herein. A GPU or other programmable device can also be an accelerator. Accelerators may be on-board the processor circuitry, in the same chip package as the processor circuitry and/or in one or more separate packages from the processor circuitry.

(95) FIG. 10 is a block diagram of another example implementation of the processor circuitry 802 of FIG. 8. In this example, the processor circuitry 802 is implemented by FPGA circuitry 1000. The FPGA circuitry 1000 can be used, for example, to perform operations that could otherwise be performed by the example microprocessor 900 of FIG. 9 executing corresponding machine-readable instructions. However, once configured, the FPGA circuitry 1000 instantiates the machine-readable instructions in hardware and, thus, can often execute the operations faster than they could be performed by a general purpose microprocessor executing the corresponding software.

(96) More specifically, in contrast to the microprocessor 900 of FIG. 9 described above (which is a general purpose device that may be programmed to execute some or all of the machine-readable instructions and/or operations represented by the flowcharts of FIGS. 4-7, but whose interconnections and logic circuitry are fixed once fabricated), the FPGA circuitry 1000 of the example of FIG. 10 includes interconnections and logic circuitry that may be configured and/or interconnected in different ways after fabrication to instantiate, for example, some or all of the machine-readable instructions and/or operations represented by the flowcharts of FIGS. 4-7. In particular, the FPGA circuitry 1000 may be thought of as an array of logic gates, interconnections, and switches. The switches can be programmed to change how the logic gates are interconnected by the interconnections, effectively forming one or more dedicated logic circuits (unless and until the FPGA circuitry 1000 is reprogrammed). The configured logic circuits enable the logic gates to cooperate in different ways to perform different operations on data received by input circuitry. Those operations may correspond to some or all of the software represented by the flowcharts of FIGS. 4-7. As such, the FPGA circuitry 1000 may be structured to effectively instantiate some or all of the machine-readable instructions of the flowcharts of FIGS. 4-7 as dedicated logic circuits to perform the operations corresponding to those software instructions in a dedicated manner analogous to an ASIC. Therefore, the FPGA circuitry 1000 may perform the operations corresponding to the some or all of the machine-readable instructions of FIGS. 4-7 faster than the general purpose microprocessor can execute the same.

(97) In the example of FIG. 10, the FPGA circuitry 1000 is structured to be programmed (and/or reprogrammed one or more times) by an end user by a hardware description language (HDL) such as Verilog. The FPGA circuitry 1000 of FIG. 10 includes example input/output (I/O) circuitry 1002 to obtain and/or output data to/from example configuration circuitry 1004 and/or external hardware (e.g., external hardware circuitry) 1006. For example, the configuration circuitry 1004 may implement interface circuitry that may obtain machine-readable instructions to configure the FPGA circuitry 1000, or portion(s) thereof. In some such examples, the configuration circuitry 1004 may obtain the machine-readable instructions from a user, a machine (e.g., hardware circuitry (e.g., programmed, or dedicated circuitry) that may implement an Artificial Intelligence/Machine Learning (AI/ML) model to generate the instructions), etc. In some examples, the external hardware 1006 may implement the microprocessor 900 of FIG. 9. The FPGA circuitry 1000 also includes an array of example logic gate circuitry 1008, a plurality of example configurable interconnections 1010, and example storage circuitry 1012. The logic gate circuitry 1008 and interconnections 1010 are configurable to instantiate one or more operations that may correspond to at least some of the machine-readable instructions of FIGS. 4-7 and/or other desired operations. The logic gate circuitry 1008 shown in FIG. 10 is fabricated in groups or blocks. Each block includes semiconductor-based electrical structures that may be configured into logic circuits. In

some examples, the electrical structures include logic gates (e.g., AND gates, OR gates, NOR gates, etc.) that provide basic building blocks for logic circuits. Electrically controllable switches (e.g., transistors) are present within each of the logic gate circuitry **1008** to enable configuration of the electrical structures and/or the logic gates to form circuits to perform desired operations. The logic gate circuitry **1008** may include other electrical structures such as look-up tables (LUTs), registers (e.g., flip-flops or latches), multiplexers, etc.

(98) The interconnections **1010** of the illustrated example are conductive pathways, traces, vias, or the like that may include electrically controllable switches (e.g., transistors) whose state can be changed by programming (e.g., using an HDL instruction language) to activate or deactivate one or more connections between one or more of the logic gate circuitry **1008** to program desired logic circuits.

(99) The storage circuitry **1012** of the illustrated example is structured to store result(s) of the one or more of the operations performed by corresponding logic gates. The storage circuitry **1012** may be implemented by registers or the like. In the illustrated example, the storage circuitry **1012** is distributed amongst the logic gate circuitry **1008** to facilitate access and increase execution speed.

(100) The example FPGA circuitry **1000** of FIG. **10** also includes example Dedicated Operations Circuitry **1014**. In this example, the Dedicated Operations Circuitry **1014** includes special purpose circuitry **1016** that may be invoked to implement commonly used functions to avoid the need to program those functions in the field. Examples of such special purpose circuitry **1016** include memory (e.g., DRAM) controller circuitry, PCIe controller circuitry, clock circuitry, transceiver circuitry, memory, and multiplier-accumulator circuitry. Other types of special purpose circuitry may be present. In some examples, the FPGA circuitry **1000** may also include example general purpose programmable circuitry **1018** such as an example CPU **1020** and/or an example DSP **1022**. Other general purpose programmable circuitry **1018** may additionally or alternatively be present such as a GPU, an XPU, etc., that can be programmed to perform other operations.

(101) Although FIGS. **9** and **10** illustrate two example implementations of the processor circuitry **802** of FIG. **8**, many other approaches are contemplated. For example, as mentioned above, modern FPGA circuitry may include an on-board CPU, such as one or more of the example CPU **1020** of FIG. **10**. Therefore, the processor circuitry **802** of FIG. **8** may additionally be implemented by combining the example microprocessor **900** of FIG. **9** and the example FPGA circuitry **1000** of FIG. **10**. In some such hybrid examples, a first portion of the machine-readable instructions represented and/or operations by the flowcharts of FIGS. **4-7** may be executed by one or more of the cores **902** of FIG. **9** and a second portion of the machine-readable instructions and/or operations represented by the flowcharts of FIGS. **4-7** may be executed by the FPGA circuitry **1000** of FIG. **10**.

(102) In some examples, the processor circuitry **802** of FIG. **8** may be in one or more packages. For example, the microprocessor **900** of FIG. **9** and/or the FPGA circuitry **1000** of FIG. **10** may be in one or more packages. In some examples, an XPU may be implemented by the processor circuitry **802** of FIG. **8**, which may be in one or more packages. For example, the XPU may include a CPU in one package, a DSP in another package, a GPU in yet another package, and an FPGA in still yet another package.

(103) From the foregoing, it will be appreciated that the above-disclosed methods and apparatus for automating shutdown processes of gas grills advantageously require a reduced degree (e.g., a reduced number) of user (e.g., human) interaction(s) to shutdown multiple (e.g., all) burners of a gas grill relative to the known shutdown processes described above that require a high degree (e.g., a high number) of user interaction(s) to shutdown multiple (e.g., all) burners of a gas grill. In some examples, automated shutdown processes disclosed herein require as few as two user interactions to shutdown multiple (e.g., all) burners of a gas grill. In other examples, automated shutdown processes disclosed herein require as few as one user interaction to shutdown multiple (e.g., all) burners of a gas grill. Implementing the automated shutdown processes disclosed herein to reduce

the degree (e.g., the number) of user interaction(s) required to shutdown multiple (e.g., all) burners of a gas grill advantageously makes the shutdown process less cumbersome in terms of the extent of user involvement that is required in connection with the shutdown process. The automated shutdown processes disclosed herein thereby provide a user experience that is improved relative to that provided by known shutdown processes for gas grills.

(104) In some examples, a grill is disclosed. In some disclosed examples, the grill includes a first burner, a second burner, a first burner valve operatively positioned between the first burner and a manifold of the grill, and a second burner valve operatively positioned between the second burner and the manifold. In some disclosed examples, the grill further comprises a controller operatively coupled to the first burner valve and the second burner valve. In some disclosed examples, the controller, in response to determining that a shutdown request has been received at the grill, is to instruct the first burner valve and the second burner valve to close.

(105) In some disclosed examples, the shutdown request is to be received via a user interface of the grill based on a user interaction with an input device of the user interface.

(106) In some disclosed examples, the shutdown request is to be received via a network interface of the grill. In some disclosed examples, the network interface is to receive the shutdown request from a remote device based on a user interaction with an input device of the remote device.

(107) In some disclosed examples, the shutdown request is to be received via an encoder of the grill. In some disclosed examples, the encoder is to receive the shutdown request based on a user interaction with a control knob of the grill. In some disclosed examples, the control knob is mechanically coupled to the encoder.

(108) In some disclosed examples, the controller is to instruct the first burner valve and the second burner valve to close in response to the controller determining that the shutdown request was received at the grill while a control system of the grill is awake.

(109) In some disclosed examples, the controller, subsequent to determining that the shutdown request has been received at the grill, and prior to instructing the first burner valve and the second burner valve to close, is to determine that a shutdown confirmation input has been received at the grill before a timer associated with a request for the shutdown confirmation input has expired.

(110) In some disclosed examples, the controller is to instruct the first burner valve and the second burner valve to concurrently close.

(111) In some disclosed examples, the controller is to instruct the first burner valve and the second burner valve to sequentially close.

(112) In some disclosed examples, the controller, in response to determining that the shutdown request has been received at the grill, is further to instruct a fuel source valve to close. In some disclosed examples, the fuel source valve is operatively positioned between a fuel source and the manifold.

(113) In some examples, a method for shutting down a grill is disclosed. In some disclosed examples, the method comprises determining, via a controller of the grill, whether a shutdown request has been received at the grill. In some disclosed examples, the method further comprises, in response to determining that the shutdown request has been received at the grill, instructing, via the controller, a first burner valve of the grill and a second burner valve of the grill to close. In some disclosed examples, the first burner valve is operatively positioned between a first burner of the grill and a manifold of the grill, and the second burner valve is operatively positioned between a second burner of the grill and the manifold.

(114) In some disclosed examples, the shutdown request is received via a user interface of the grill based on a user interaction with an input device of the user interface.

(115) In some disclosed examples, the shutdown request is received via a network interface of the grill. In some disclosed examples, the network interface receives the shutdown request from a remote device based on a user interaction with an input device of the remote device.

(116) In some disclosed examples, the shutdown request is received via an encoder of the grill. In

some disclosed examples, the encoder receives the shutdown request based on a user interaction with a control knob of the grill. In some disclosed examples, the control knob is mechanically coupled to the encoder.

(117) In some disclosed examples, instructing the first burner valve and the second burner valve to close occurs in response to determining, via the controller, that the shutdown request was received at the grill while a control system of the grill is awake.

(118) In some disclosed examples, the method further comprises, subsequent to determining that the shutdown request has been received at the grill, and prior to instructing the first burner valve and the second burner valve to close, determining, via the controller, that a shutdown confirmation input has been received at the grill before a timer associated with a request for the shutdown confirmation input has expired.

(119) In some disclosed examples, instructing the first burner valve and the second burner valve to close includes instructing the first burner valve and the second burner valve to concurrently close.

(120) In some disclosed examples, instructing the first burner valve and the second burner valve to close includes instructing the first burner valve and the second burner valve to sequentially close.

(121) In some disclosed examples, the method further comprises, in response to determining that the shutdown request has been received at the grill, instructing, via the controller, a fuel source valve to close. In some disclosed examples, the fuel source valve is operatively positioned between a fuel source and the manifold.

(122) In some examples, a non-transitory computer-readable medium comprising computer-readable instructions is disclosed. In some disclosed examples, the computer-readable instructions, when executed, cause one or more processors of a grill to determine whether a shutdown request has been received at the grill. In some disclosed examples, the computer-readable instructions, when executed, cause the one or more processors, in response to determining that the shutdown request has been received at the grill, to instruct a first burner valve of the grill and a second burner valve of the grill to close. In some disclosed examples, the first burner valve is operatively positioned between a first burner of the grill and a manifold of the grill, and the second burner valve is operatively positioned between a second burner of the grill and the manifold.

(123) In some disclosed examples, the shutdown request is received via a user interface of the grill based on a user interaction with an input device of the user interface.

(124) In some disclosed examples, the shutdown request is received via a network interface of the grill. In some disclosed examples, the network interface receives the shutdown request from a remote device based on a user interaction with an input device of the remote device.

(125) In some disclosed examples, the shutdown request is received via an encoder of the grill. In some disclosed examples, the encoder of the grill receives the shutdown request based on a user interaction with a control knob of the grill. In some disclosed examples, the control knob is mechanically coupled to the encoder.

(126) In some disclosed examples, the computer-readable instructions, when executed, cause the one or more processors to instruct the first burner valve and the second burner valve to close in response to determining, via the one or more processors, that the shutdown request was received at the grill while a control system of the grill is awake.

(127) In some disclosed examples, the computer-readable instructions, when executed, further cause the one or more processors, subsequent to determining that the shutdown request has been received at the grill, and prior to instructing the first burner valve and the second burner valve to close, to determine that a shutdown confirmation input has been received at the grill before a timer associated with a request for the shutdown confirmation input has expired.

(128) In some disclosed examples, the computer-readable instructions, when executed, cause the one or more processors to instruct the first burner valve and the second burner valve to concurrently close.

(129) In some disclosed examples, the computer-readable instructions, when executed, cause the

one or more processors to instruct the first burner valve and the second burner valve to sequentially close.

(130) In some disclosed examples, the computer-readable instructions, when executed, further cause the one or more processors, in response to determining that the shutdown request has been received at the grill, to instruct a fuel source valve to close. In some disclosed examples, the fuel source valve is operatively positioned between a fuel source and the manifold.

(131) Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

(132) The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Claims

1. A grill, comprising: a first burner; a second burner; a first burner valve operatively positioned between the first burner and a manifold of the grill; a second burner valve operatively positioned between the second burner and the manifold; and a controller operatively coupled to the first burner valve and the second burner valve, wherein the controller, in response to determining that a shutdown request has been received at the grill, is to; initiate a timer associated with a shutdown confirmation request, the timer providing a time period by which a shutdown confirmation input must be received at the grill in response to the shutdown confirmation request; determine whether the shutdown confirmation input has been received at the grill before the time period has expired; and in response to determining that the shutdown confirmation input has been received at the grill before the time period has expired, instruct the first burner valve and the second burner valve to close.
2. The grill of claim 1, wherein the shutdown request is to be received via a user interface of the grill based on a user interaction with an input device of the user interface.
3. The grill of claim 1, wherein the shutdown request is to be received via a network interface of the grill, and wherein the network interface is to receive the shutdown request from a remote device based on a user interaction with an input device of the remote device.
4. The grill of claim 1, wherein the shutdown request is to be received via an encoder of the grill, and wherein the encoder is to receive the shutdown request based on a user interaction with a control knob of the grill, the control knob being mechanically coupled to the encoder.
5. The grill of claim 1, wherein the controller is to instruct the first burner valve and the second burner valve to concurrently close.
6. The grill of claim 1, wherein the controller is to instruct the first burner valve and the second burner valve to sequentially close.
7. The grill of claim 1, wherein the controller, in response to determining that the shutdown confirmation request has been received at the grill before the time period has expired, is further to instruct a fuel source valve to close, the fuel source valve being operatively positioned between a fuel source and the manifold.
8. A method for shutting down a grill, the method comprising: determining, via a controller of the grill, whether a shutdown request has been received at the grill; and in response to determining that the shutdown request has been received at the grill: initiating, via the controller, a timer associated with a shutdown confirmation request, the timer providing a time period by which a shutdown confirmation input must be received at the grill in response to the shutdown confirmation request; determining, via the controller, whether the shutdown confirmation input has been received at the grill before the time period has expired; and in response to determining that the shutdown confirmation input has been received at the grill before the time period has expired, instructing, via

the controller, a first burner valve of the grill and a second burner valve of the grill to close, the first burner valve being operatively positioned between a first burner of the grill and a manifold of the grill, the second burner valve being operatively positioned between a second burner of the grill and the manifold.

9. The method of claim 8, wherein instructing the first burner valve and the second burner valve to close includes instructing the first burner valve and the second burner valve to concurrently close.

10. The method of claim 8, wherein instructing the first burner valve and the second burner valve to close includes instructing the first burner valve and the second burner valve to sequentially close.

11. The method of claim 8, further comprising, in response to determining that the shutdown confirmation request has been received at the grill before the time period has expired, instructing, via the controller, a fuel source valve to close, the fuel source valve being operatively positioned between a fuel source and the manifold.

12. A non-transitory computer-readable medium comprising computer-readable instructions that, when executed, cause one or more processors of a grill to at least: determine whether a shutdown request has been received at the grill; and in response to determining that the shutdown request has been received at the grill: initiate a timer associated with a shutdown confirmation request, the timer providing a time period by which a shutdown confirmation input must be received at the grill in response to the shutdown confirmation request; determine whether the shutdown confirmation input has been received at the grill before the time period has expired; and in response to determining that the shutdown confirmation input has been received at the grill before the time period has expired, instruct a first burner valve of the grill and a second burner valve of the grill to close, the first burner valve being operatively positioned between a first burner of the grill and a manifold of the grill, the second burner valve being operatively positioned between a second burner of the grill and the manifold.

13. The non-transitory computer-readable medium of claim 12, wherein the computer-readable instructions, when executed, cause the one or more processors to instruct the first burner valve and the second burner valve to concurrently close.

14. The non-transitory computer-readable medium of claim 12, wherein the computer-readable instructions, when executed, cause the one or more processors to instruct the first burner valve and the second burner valve to sequentially close.

15. The grill of claim 1, wherein the controller, subsequent to instructing the first burner valve and the second burner valve to close, is to determine, via flame detection circuitry of the grill, whether the first burner and the second burner have been extinguished.

16. The grill of claim 15, wherein in response to determining that at least one of the first burner and the second burner has not been extinguished, the controller is to re-instruct the at least one of the first burner valve and the second burner valve to close.

17. The method of claim 8, further comprising, subsequent to instructing the first burner valve and the second burner valve to close, determining, via flame detection circuitry of the grill, whether the first burner and the second burner have been extinguished.

18. The method of claim 17, further comprising, in response to determining that at least one of the first burner and the second burner has not been extinguished, re-instructing, via the controller, the at least one of the first burner valve and the second burner valve to close.

19. The non-transitory computer-readable medium of claim 12, wherein the computer-readable instructions, when executed, cause the one or more processors, subsequent to instructing the first burner valve and the second burner valve to close, to determine, via flame detection circuitry of the grill, whether the first burner and the second burner have been extinguished.

20. The non-transitory computer-readable medium of claim 19, wherein the computer-readable instructions, when executed, cause the one or more processors, in response to determining that at least one of the first burner and the second burner has not been extinguished, to re-instruct the at least one of the first burner valve and the second burner valve to close.

